



Tribhuvan University

Faculty of Humanities and Social Science

Semester: IV

Subject: Database Management System

2021 Batch

Group B

Attempt any SIX questions

2. Define database system. Write the merits and demerits of DBMS.
3. Draw and explain three schema architecture of the database system.
4. What are an entity and its attributes? Write the difference between a strong and weak entity set with an example.
5. Why integrity constraints are used in databases? Explain different types of integrity constraints with examples.
6. What is functional dependency? Explain its types with examples.
7. Differentiate between discretionary access control and mandatory access control with example.
8. Write the ACID properties of transactions. Explain the concepts of serializability in database transactions.

Group C

Attempt any TWO questions.

9. Why normalization is required in database? Explain the 1NF, 2NF and 3NF of normalization with example.
10. How participation constraints are used in ER diagram? Explain. Also Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period and has an associated due date, and the date when the payment was received.
11. Consider the following employee database, where primary keys are underlined
Supplier (supplier-id, supplier-name, city)
Suppliers (Supplier-id, parts-id, quantity)
Parts (part-id, par-name, color, weight)
a) Write a SQL query to create above table.
b) Find the name of all suppliers who has supplied parts with quantity more than
c) Find the name of all parts supplied by "RD Traders"
d) Find the number of parts supplied by "S02"
e) Find the name of all suppliers located in city "KTM".

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2020 Batch

Group B

Attempt any SIX questions

2. Define database and DBMS. Explain the benefits of DBMS in detail.
3. What are different SQL constraints to specify the rules for the data in the database? Explain.
4. What is relational algebra? Explain different RA operations.
5. How candidate key is different from primary key? Also explain foreign key in detail.
6. What is a stored procedure? How can it be used in a database? Explain with an example.
7. What are the main purposes of concurrency control? Explain two phase locking protocol in detail.
8. What is database security? Describe steps of query processing.

Group C

Attempt any TWO questions.

9. What are fully and partial functional dependencies? Explain 2NF and 3NF in detail.
10. How different join operations can be performed? Explain with proper examples. Consider a database of a shipment company. Shipped items are received into the system at a single retail center. Shipped items make their way to their destination via one or more standard transportation events (ie, flights, truck deliveries). The transportation events follow a regular schedule. Create an ER diagram to represent this information.
11. Consider the following tables:

Book(Bookid, Title, Publisher, Published Date, Price)

Book_Authors(Bookid, Author name)

Borrower(Card No, Name, Address, Contact)

Bookissue(Bookid, CardNo, IssuedDate, Due Date)

Write SQL query for the following:

- a) Find the name of the books taken by Ramesh Sharma.
- b) Find the Issued book name and the borrower's name who have Issued date before 15th May, 2023 and due date after 15th June, 2023.
- c) Find the borrower's name who has taken the book written by Laxmi Prashad Devkota.
- d) Find the number of books issued on the date 25th April, 2023
- e) Find the borrower's name who has taken the book having price more than average price.

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2019 Batch

Group B

Attempt any SIX questions

2. What is DBMS? Describe the advantages and disadvantages of using DBMS.
3. Describe 3-schema architecture of database.
4. What are roles and responsibilities of DBA.
5. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.
6. What are the states of transaction model? Describe the ACID properties of transaction.
7. Describe the use of ER diagram. Draw an ER diagram for a university to design a database for it.
8. What is DDL and DML queries?

Consider the following relation.

Customer name	Loan number
X	L01
Y	L02

Relation borrower

Loan number	Branch name	Amount
L01	B1	5000
L02	B2	6000

Relation loan

Write a relational algebra expression to find all customer who taken loan from branch "B1".

Group C

Attempt any TWO questions.

9. Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples.
10. Differentiate between stored procedure and trigger. Why concurrency control is needed? Describe timestamp based concurrency protocol.
11. What is data model? What are its types? Describe the structure of database.

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Semester: IV
Subject: Database Management System
2018 Batch

Group B

Attempt any SIX questions

2. What is database? Discuss the importance of DBMS.
3. Define union compatibility. Given following relations, show the resulting relation of Director $\cap$ Actor and Actor $\cup$ Director?

Director

Fname	Lname
Deepak	Giri
Bhuwan	KC
Dipendra	Khanel

Actor

First name	Last name
Deepak	Giri
Rajesh	Hamal
Dipendra	Khanel

4. How Order By and Group BY Having Clause in SQL are different? Illustrate with suitable examples.
5. Why normalization is needed? When any relation is said to be in 1NF and 2NF?
6. How cost of query is measured during query processing? Construct query expression tree for the relational algebra query;
TLFName LName, Address(ODName="Research" (Department Dnumber=Dno (Employee * Works_on
7. How wait-for-graph is constructed? How can you use it to detect deadlock in a schedule of transactions? Support your answer with an example.
8. How read and write operations are performed in a transaction? List the various states where transactions can enter into.

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Group C

Attempt any TWO questions.

9. Given the schema as below, write relational algebra and SQL queries for following;

Sailors(sid, sname, rating, age)

Boats(bid, bname, colour)

Reserves(sid, bid, reservedate)

- a) Find name, rating and age of all the sailors.
 - b) Find the name and color of boats having color blue.
 - c) Find name of the boats reserved on date 2021/07/20.
 - d) Find the names of boats and sailors who have reserved boats having color green.
 - e) Use Left and Right Outer Join between Sailors and Reserves.
10. a) Design an ER diagram of your choice. The ER diagram should contain at least five entities. Out of those entities, one should be weak entity. Your entities should have appropriate attributes including primary key attributes, if any. There should be presence of one to one relationship, one to many relationship and many to many relationships in your ER diagram.
- b) Convert the ER diagram in Q. 19. a. into equivalent relational tables.
11. Define stored procedures and triggers. How they are applicable in database? Illustrate with examples, how can you create before and after triggers on INSERT query.

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Tribhuvan University
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Semester: IV
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2017 Batch

Group B

Attempt any SIX questions

2. What is DBMS? Describe the merits of using DBMS.
3. What is Data Model? What is the difference between hierarchical and network data model? what are its types?
4. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.
5. Discuss about materialized and pipelined evaluation of query expression in query optimization
6. Explain with examples, how lost update and dirty read problems can occur in sentence of transactions?
7. How time stamp ordering protocol is used to ensure concurrency control?
8. What is the advantage and Disadvantage of using stored procedure? How can you create and execute stored procedure?

Group C

Attempt any TWO questions.

9. Design an ER diagram for Hospital System. Use your assumption for the selection attributes, entities and relationships. Show the use of total and partial participation along with the appropriate cardinalities,
10. Discuss the importance of normalization in DBMS, Describe 1NF, 2NF and 3NF with examples.
11. Consider a database system with following schemes:
Restaurant (Rname, Rlocation, Fname)
Cook (Cname, Cspeciality)
Worksat (Cname, Rname, Workinghrs, Shift)
Food (Fname, Cname, Catagory)

Now write SQL statements and relational algebra statements for following Queries:

- a) Select the name and location of al restaurants.
- b) Find the working hours of cool named "Sita".
- c) Select name of the foods cooked by "Ramesh".
- d) Use join to select name of restaurants where food of category "breakfast" is available.
- e) Find the name of cooks who work as "KFC"

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